Communication Design for Multi-Agent Aircraft Formations Under Attack

Ali Eslami and Jiangbo Yu

***Abstract*— Aircraft in close formation exchange navigation data over communication links and interact physically through their wakes, the disturbed air they leave behind. If an attacker corrupts data on one link, the receiving aircraft's response can propagate through both communication and wake interactions. We ask how many communication links a formation needs to minimize the worst-case steady-state attack gain. Given the local controllers, the designer chooses the communication links and their weights. With knowledge of the design, the attacker selects one active link and a bounded-energy signal. Under the stated steady-state response conditions, we derive a lower bound on attack amplification that holds across formation geometries sharing the same physical graph. We develop a rule for selecting links and weights and a test that certifies whether the design attains the bound at steady state. For ordered formations that weight all pairwise errors equally, the minimum number of communication links needed to attain the bound is $2N-3$ for N aircraft. Case studies with delayed wake coupling show that the proposed design can approach the analytical bound and that formation geometry affects the dynamic response to attacks.**

I. Introduction

Aircraft flying in close formation can reduce their energy consumption by using the upwash generated by preceding aircraft [1]. Each aircraft acts as an agent with its own local controller, and each follower uses navigation data received over communication links to maintain its position in the formation. Aircraft also interact through their wakes, the disturbed air they leave behind. These wakes change the aerodynamic forces on downstream aircraft. If an attacker corrupts data on one link, the error first changes the receiving aircraft's command and can then spread through both communication and wake interactions.

Given the role of wake interactions and information exchange in close formation, prior work has examined how disturbances and attacks affect formation control and how the communication graph shapes the response. The authors in [1] studied disturbance amplification in aircraft coupled through wakes and designed controllers for string stability. The authors in [2] addressed wake interference and deception attacks together through event-triggered formation control. In the related setting of vehicle platoons, the authors in [3] examined which attacks on communications and local sensors can evade detection. The importance of the communication graph for formation stability was established in [4]. The authors in [5] then showed that adding a link in directed leader–follower consensus can increase the disturbance-to-state $\mathcal{H}_\infty$ gain. Furthermore, the authors in [6] linked platoon connectivity to disturbance rejection, attack detection, and resilient distributed algorithms. For linear consensus networks, the authors in [7] derived limits linking the number of network connections to performance under stochastic disturbances. The authors in [8] and [9] took a game-theoretic approach, studying which agents an attacker selects for signal injection and where a defender chooses to add self-feedback.

These findings motivate treating the communication graph as a design variable. The authors in [10] related platoon robustness to topology and synthesized distributed controllers. The authors in [11] selected leaders to reduce error under link noise. The authors in [12] designed sparse relative sensing networks under robust performance constraints. The authors in [13] designed sparse feedback gains by penalizing communication links while optimizing $\mathcal{H}_2$ performance, and the authors in [14] studied edge selection to improve consensus coherence under stochastic disturbances. The authors in [15] designed edge weights for a fixed graph with disturbances entering the links. The authors in [16] studied edge addition to increase the smallest grounded-Laplacian eigenvalue.

These studies explain how wake interactions affect aircraft formation control and how communication topology shapes robustness to disturbances and attacks. However, they do not establish how many communication links a physically coupled multi-agent aircraft formation needs to minimize the worst-case steady-state amplification of a single-link attack, or which links and weights attain that minimum. We address this question and examine how wake coupling at different formation geometries affects the dynamic response.

To address this gap, we formulate a minimax problem in which communication links and their weights are chosen to minimize the worst-case amplification of a single-link attack. Knowing the selected links and weights, the attacker chooses one active link and a bounded-energy signal to inject into its data. To isolate the effect of communication design, we keep the local controllers fixed, normalize each follower's incoming weights, and measure relative position and velocity errors over the same aircraft pairs. We then use the steady-state response to derive a lower bound on attack amplification and construct communication designs that can attain it. Finally, we evaluate these designs under delayed wake coupling to examine how formation geometry affects their performance beyond steady state.

The main contributions of this paper are therefore as follows:

1) We derive a lower bound on attack amplification that holds across formation geometries sharing the same

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physical graph.

2) We select communication links and weights and provide a test for whether the resulting design attains the lower bound on steady-state attack amplification.
3) We bound the smallest attack amplification achievable with a given number of communication links. We also determine exact minimum link counts for attaining the steady-state limit for complete and tree-shaped physical graphs.

II. Formation Model and Attack Problem

A. Aircraft Dynamics

Consider a multi-agent formation of $N \geq 2$ aircraft. Agent 0 is the leader, and agents $1, \ldots, N-1$ are the followers. We linearize the aircraft and their local controllers about steady cruise in formation, including the nominal wake loads. We express the states in deviation coordinates relative to the nominal cruise trajectories. The leader is assumed to stay on its nominal trajectory, giving $x_0(t) = 0$ in these coordinates.

The dynamics of each follower $i \in \{1, \ldots, N-1\}$ are governed by

$$\dot{x}_i = A_i^p x_i + B_i u_i + \sum_{j \in \mathcal{N}_i^p} P_{ji} x_j(t - \tau_{ji}). \tag{1}$$

The matrix A_i^p describes the local dynamics of follower i, including how its own motion changes the wake load it experiences. The term $B_i u_i$ gives the effect of the cooperative command. The sum captures the wake coupling from upstream aircraft $j \in \mathcal{N}_i^p$, with P_{ji} mapping each delayed upstream state into the follower dynamics and τ_{ji} the wake travel time.

Figure 1 shows the five-aircraft formation and the alternate heights of its followers. The arrows distinguish the physical coupling through wakes from the communication links that carry data to the follower controllers. At fixed aircraft positions, changing the communication links leaves the wake interactions unchanged.

B. Cooperative Control and Attack Model

The aircraft output shared with other aircraft is

$$y_i = C_i x_i = (p_i^T, v_i^T, \chi_i)^T \in \mathbb{R}^7, \tag{2}$$

where p_i and v_i are the three-dimensional position and velocity deviations from the prescribed flight condition, and χ_i is the course deviation. The cooperative control law is

$$u_i = K_F \sum_j w_{ij}(\widetilde{y}_{ji} - y_i), \quad w_{ij} \geq 0, \quad \sum_j w_{ij} = 1. \tag{3}$$

Here $\widetilde{y}_{ji}$ is the output received by aircraft i from aircraft j. Communication is modeled without delay, so $\widetilde{y}_{ji}(t) = y_j(t)$ on an uncorrupted link. A positive weight w_{ij} selects the communication link $j \to i$. The weight matrix W has a zero diagonal and a zero leader row. The unit row sums at the followers keep the local feedback term equal to $-K_F y_i$ as we vary the links and weights.

The attacker selects one active link $j_* \to i_*$ after the design is known and keeps that link fixed throughout the attack. It adds the attack signal $S_a a(t)$ to the data received on that link,

$$\widetilde{y}_{j_* i_*} = y_{j_*} + S_a a(t), \qquad \int_0^\infty a(t)^T a(t)\, dt \leq 1. \tag{4}$$

The normalized attack input $a(t) \in \mathbb{R}^7$ specifies changes to the three position components, three velocity components, and course. The diagonal matrix S_a converts the input to the transmitted units. The attacker may choose any waveform satisfying (4), including a pulse or a sinusoid applied over a finite interval. The energy budget is shared across the seven components of the attack input.

Writing $A_i = A_i^p - B_i K_F C_i$, substitution gives

$$\begin{aligned} \dot{x}_i = A_i x_i + B_i K_F \sum_j w_{ij} y_j + \sum_{j \in \mathcal{N}_i^p} P_{ji} x_j(t - \tau_{ji}) \\ + \mathbf{1}_{\{i = i_*\}} w_{i_* j_*} B_i K_F S_a a. \end{aligned} \tag{5}$$

We select K_F so that the local closed-loop matrices A_i are Hurwitz for the formations under consideration, and hold it fixed when designing W. Every follower must be reachable from the leader through communication. The wake interactions and communication links must also admit a common forward order. This makes the formation a cascade of stable local systems coupled by feedforward paths, including the delayed wake paths.

C. Performance Criterion and Link Budget

We measure formation keeping by the errors in relative position and velocity across the physical pairs. For a pair (j, i), the vector $p_i - p_j$ is the deviation from the prescribed relative position, and $v_i - v_j$ is the relative velocity error. Let $\mathcal{E}_p$ be the fixed physical edge set, with $Q = |\mathcal{E}_p|$ and a connected underlying undirected graph. We collect these errors in the performance output

$$z = \frac{1}{\sqrt{Q}} \operatorname{col}_{(j,i) \in \mathcal{E}_p} W_F \begin{bmatrix} p_i - p_j \\ v_i - v_j \end{bmatrix}, \tag{6}$$

where W_F scales the position and velocity errors and col stacks the resulting vectors. With $1/\sqrt{Q}$, $\|z(t)\|_2$ is the root mean square of the scaled error norms across the Q pairs. We keep the same pairs and scales when comparing communication designs and geometries.

Starting from undisturbed flight, we ask how much formation error an attacker can produce with one unit of input energy. Let q denote the formation geometry, specified by the nominal positions of the followers relative to the leader. It determines the linearized aircraft and wake model, including the wake delays. For a fixed link $j \to i$, let $T_{ji}(s; q, W)$ be the transfer from the normalized attack input a to the performance output z. Its $\mathcal{H}_\infty$ norm is the supremum of $\|z\|_{L_2}$ over the waveforms satisfying (4). The L_2 norm measures the error over the entire response. The attacker can choose any active link, so the worst single-link gain is

$$\Gamma(q, W) = \max_{w_{ij} > 0} \|T_{ji}(\cdot; q, W)\|_\infty, \tag{7}$$

where $\|T\|_\infty = \sup_{\omega \geq 0} \bar{\sigma}(T(\mathrm{i}\omega))$. Thus Γ measures the worst amplification over both link choice and attack waveform.

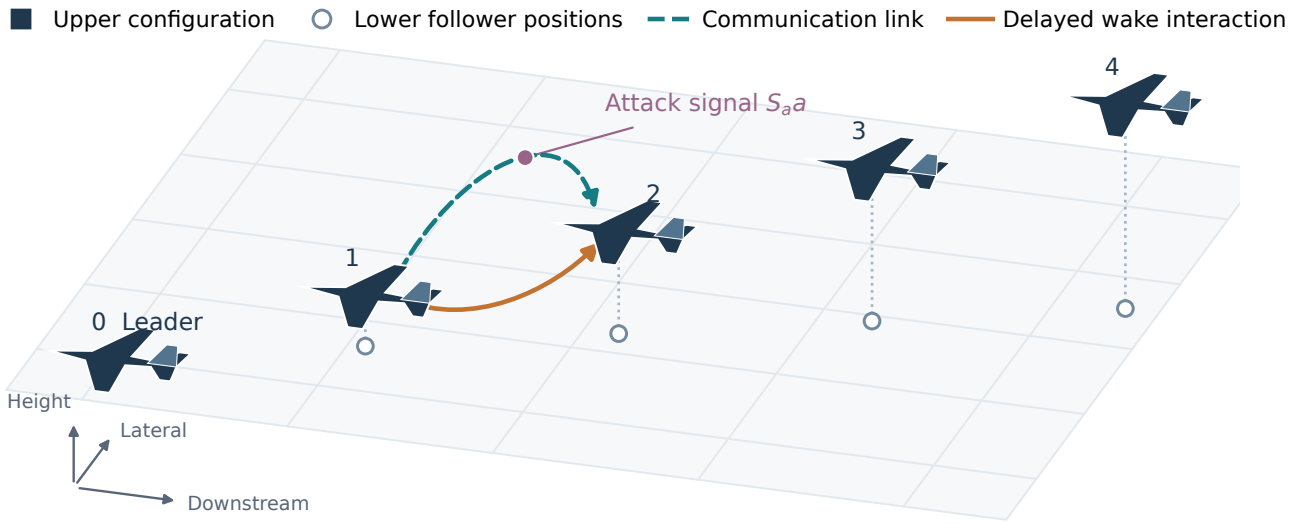


Fig. 1. Five-aircraft formation. The dashed arrow shows the communication link from aircraft 1 to aircraft 2, and the solid arrow shows the wake interaction between them.

For input energy E_a, the supremum of the output L_2 norm is $\sqrt{E_a}\,\Gamma$. The designer can activate at most L communication links, including links from the leader. Let $\mathcal{W}_L$ contain the weight matrices satisfying this budget and the preceding graph constraints. At a given geometry q, our goal is to bound and minimize $\Gamma(q, W)$ over $W \in \mathcal{W}_L$.

D. Steady-State Response

The gain at zero frequency gives a lower bound on the worst-case gain. To compute it, we describe how a follower settles when its incoming data or wake load has a constant error. The following assumption gives the local responses used in Section III.

Assumption 1: For every follower and eligible geometry, the steady-state response satisfies

$$\begin{aligned} C_i(-A_i)^{-1}B_iK_F &= H_0, \\ C_i(-A_i)^{-1}P_{ji} &= 0, \end{aligned} \qquad H_0^2 = H_0, \tag{8}$$

with the same matrix H_0. The wake identity holds for each $j \in \mathcal{N}_i^p$.

The first identity gives the same local response H_0 to a constant offset in the weighted incoming data at every follower and geometry. The second says that a constant wake load alone leaves no lasting error in the shared output. The identity $H_0^2 = H_0$ says that applying the map to one of its own outputs preserves that output.

In the aircraft model, integral action balances constant wake loads at equilibrium. A position bias in the incoming data shifts the position reference, so it can persist as a position offset after the aircraft settles. The settled output has zero velocity and course deviations, and H_0 preserves such a position offset. Section IV verifies these identities for the aircraft model.

III. Steady-State Bounds and Link Design

A. Attack Propagation and Gain Bound

We first calculate the settled formation error caused by an attack on a link entering one follower. This gives a lower bound on the full attack gain, since the full gain includes the response at zero frequency.

Under Assumption 1, the attack signal contributes directly to the settled output of the receiving agent. Every follower that receives this output inherits a copy scaled by its incoming link weight. Consider one component of the settled output. Collect the direct responses to the attack in d and the total settled responses in r. The leader has zero output deviation and receives no packets. The follower balance therefore uses only the follower rows and columns W_f of W, giving $r = d + W_f r$. The receiver's direct response to an attack on a leader link also appears in d. Solving for r gives $r = Rd$, where

$$R = (I - W_f)^{-1} = \sum_{\ell=0}^{N-2} W_f^{\ell}. \tag{9}$$

The identity term gives the direct response. Each additional power gives the part carried by paths with one more link. For example, the path $i \to k \to m$ contributes the product $w_{ki}w_{mk}$. The sum ends at $N-2$ because there are $N-1$ followers and no directed cycles.

To score formation keeping, we take the difference of the settled responses for each physical pair. The matrix H collects these differences, one per row, using zero for the leader's response. Thus HRe_i gives the physical differences produced by a unit response injected at follower i, where e_i is its coordinate vector. The matrix $J = W_F[H_0]_{1:6,:}S_a$ gives the scaled local position and velocity response to the normalized attack input for a unit link weight. Its largest singular value is $\beta = \|J\|_2 > 0$.

The earliest follower in the common forward order receives only from the leader. This link carries weight one. An attack

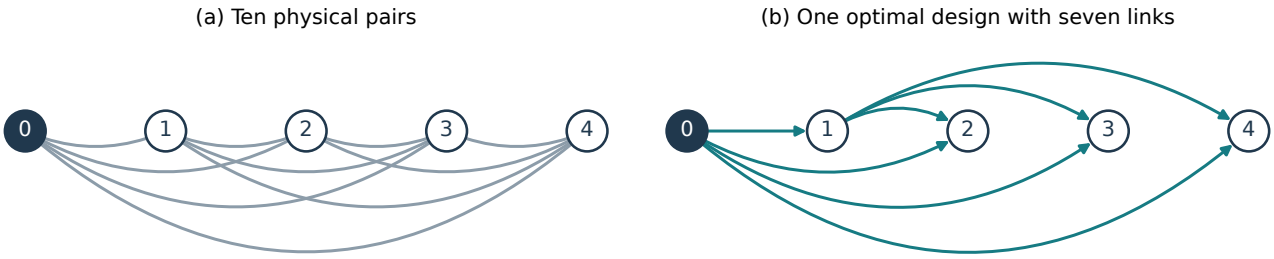


Fig. 2. Physical pairs and communication links for five aircraft. (a) All ten pairs contribute to the formation error in equation (6). (b) An optimal steady-state design given by equation (19), with the minimum of seven links.

on it therefore creates a local response that the communication design must accommodate. To bound the error this response creates, give follower i a unit scalar response and allow all other followers to take whatever values minimize the squared physical differences. With $K = H^T H \succ 0$, this minimum is

$$\rho_i = \min_{g_i=1} \|Hg\|_2^2 = \frac{1}{(K^{-1})_{ii}}. \tag{10}$$

Here g collects the scalar follower responses, and ρ_i is the grounded effective conductance [17]. Only followers without an upstream follower in the physical graph can be first. Let $\mathcal{F}$ be this set, and put $\rho_* = \min_{i\in\mathcal{F}} \rho_i$. This accounts for the choice of which follower is first.

Theorem 1: For every admissible design, the steady-state link transfer is

$$T_{ji}(0) = \frac{w_{ij}}{\sqrt{Q}}(HRe_i) \otimes J, \tag{11}$$

where $\otimes$ denotes the Kronecker product. With $\alpha_i = \max_j w_{ij}$, the worst single-link steady-state gain Γ_{ss} satisfies

$$\begin{aligned} \Gamma(q, W) \geq \Gamma_{\mathrm{ss}}(W) &:= \frac{\beta}{\sqrt{Q}} \max_i \alpha_i \|HRe_i\|_2 \\ &\geq \beta\sqrt{\frac{\rho_*}{Q}}. \end{aligned} \tag{12}$$

Both lower bounds are independent of formation coordinates.

Proof: At zero frequency, (8) eliminates the wake terms from the shared outputs. The attack signal on $j \to i$ contributes $w_{ij}H_0S_a a$ to the output of follower i. Every follower output lies in the range of H_0, so applying H_0 to an incoming follower output preserves it. The settled response thus follows the balance in (9), with a direct attack contribution only at i. Taking the physical differences and applying the output scales gives (11). The Kronecker norm identity and $\|T(0)\|_2 \leq \|T\|_\infty$ give the first inequality in (12).

The first follower in any combined order belongs to $\mathcal{F}$ and receives only from the leader, with weight one. For this follower i, acyclicity gives $(Re_i)_i = 1$, hence $\|HRe_i\|_2^2 \geq \rho_i \geq \rho_*$. Its mandatory leader link proves the final inequality. ∎

A pulse in the worst steady-state attack direction can approach Γ_{ss} as its duration T increases, with amplitude scaled by $1/\sqrt{T}$ to keep its energy fixed.

B. Communication Weight Design

Choose a follower $i \in \mathcal{F}$ with $\rho_i = \rho_*$ and set

$$h = \frac{K^{-1}e_i}{(K^{-1})_{ii}}, \qquad Kh = \rho_i e_i. \tag{13}$$

The vector h minimizes the squared physical error subject to $h_i = 1$. The maximum principle for a grounded Laplacian gives $0 \leq h_j \leq 1$ for each follower j. Assume the leader can transmit to all followers and follower i can transmit to the other followers. Since $i \in \mathcal{F}$, the physical graph admits a forward order with the leader and i first. These communication links preserve that order.

Lemma 1: Set $w_{i0} = 1$ and, for every other follower j, choose

$$w_{ji} = h_j, \qquad w_{j0} = 1 - h_j. \tag{14}$$

Set all remaining weights to zero. This uses at most $2N - 3$ links and has

$$\Gamma_{\mathrm{ss}}^2 = \frac{\beta^2}{Q} \max\left\{\rho_i, \max_{j\neq i} K_{jj} \max(h_j, 1-h_j)^2\right\}. \tag{15}$$

The inner maximum is over the other followers. If all its terms are at most ρ_i, the design is globally steady-state optimal at every budget that permits it. For $N = 2$, only the term ρ_i remains.

Proof: Column i of R is h, and every other column j is e_j. Hence the link entering i has squared propagation norm $h^T Kh = \rho_i$. A link from k to another follower j has squared propagation norm $K_{jj}w_{jk}^2$. Theorem 1 gives (15). Meeting its lower bound proves global optimality. ∎

The construction gives an attack on the leader-to-i link the displacement pattern with least physical error. To attain the bound, the gain on every other link must be at most the gain on that link.

Once follower i is selected, one linear solve gives the weights and $N - 2$ scalar inequalities test optimality. Each other follower j gives weight h_j to i and the rest to the leader. A restricted communication range can prevent the use of these links.

C. Link-Budget Bounds

An attack entering a follower cannot affect earlier aircraft in the common forward order. We can use this restriction to strengthen the gain bound when few communication links are

available. Fix an admissible order and let d_i be the number of links entering follower i, including a possible leader link. Define

$$\sigma_i = \min_{g_i=1,\ g_j=0 \text{ for earlier followers } j} g^T K g. \tag{16}$$

This is the least squared physical error compatible with a unit response at i and zero response at all earlier followers.

Proposition 1: For every admissible design in this order,

$$\Gamma_{\mathrm{ss}}^2(W) \geq \frac{\beta^2}{Q} \max_i \frac{\sigma_i}{d_i^2}. \tag{17}$$

Proof: The propagation vector Re_i satisfies the constraints in (16). Since the d_i positive incoming weights sum to one, $\alpha_i \geq 1/d_i$. Substitution in (12) proves the claim. ■

Minimizing the right-hand side over positive integer counts with $\sum_i d_i \leq L$ gives a budget-dependent lower bound. Each count is at most the number of earlier aircraft available as senders. If several forward orders are allowed, take the minimum over those orders as well.

D. Exact Budgets for Complete Graphs and Trees

Consider the ordered physical graph containing every edge $j \to i$ with $0 \leq j < i \leq N-1$. Every pairwise error has equal weight, so

$$Q = N(N-1)/2, \qquad K = NI_{N-1} - \mathbf{1}\mathbf{1}^T. \tag{18}$$

Figure 2 shows the five-aircraft case. The performance output includes all ten physical pairs, and the design attaining the steady-state bound uses seven communication links. The physical order is fixed, and communication may use any forward link. Let $b_L = \min_{W \in \mathcal{W}_L} \Gamma_{\mathrm{ss}}(W)$.

Theorem 2: For every $N \geq 2$, the bound $b_L \geq \beta/\sqrt{N-1}$ is attained if and only if $L \geq 2N-3$, assuming $L \geq N-1$. An attaining design is

$$w_{10} = 1, \qquad w_{i0} = w_{i1} = 1/2 \quad (i = 2, \ldots, N-1). \tag{19}$$

For $N-1 \leq L < 2N-3$, set $r = 2(N-1) - L$. Every admissible design satisfies the stricter bound

$$\Gamma_{\mathrm{ss}}(W) \geq \beta \sqrt{\frac{2r}{(N-1)(r+1)}} > \frac{\beta}{\sqrt{N-1}}. \tag{20}$$

At $L = N-1$, the leader-to-all design attains $b_L = \beta\sqrt{2/N}$. Sharpness of (20) at intermediate budgets is not claimed.

Proof: Here $\mathcal{F} = \{1\}$, so Lemma 1 selects $i = 1$. Equation (18) gives $\rho_1 = N/2$, $h_1 = 1$, and $h_j = 1/2$ for $j \geq 2$. Since $K_{jj} = N-1$, all terms in the inner maximum in (15) equal $(N-1)/4 < N/2$. Thus the design attains the bound with $2N-3$ links.

For this graph, the constrained minimum in (16) is

$$\sigma_i = \frac{Ni}{i+1}. \tag{21}$$

Indeed, minimizing over the later components gives the common value $1/(i+1)$. If $n = N-1$ receivers use at most L links, at least $r = 2n - L$ have only one parent. One of these has index $i \geq r$, and Proposition 1 gives $\Gamma_{\mathrm{ss}}^2 \geq (\beta^2/Q)Nr/(r+1)$. This proves (20), which strictly exceeds the floor when $r \geq 2$. At $L = N-1$, the leader-to-all design has $R = I$ and squared gain $\beta^2(N-1)/Q$, attaining the stated bound. ■

Only a linear number of communication links is needed although the physical output contains a quadratic number of pairs. Additional links cannot improve this steady-state optimum, but may affect dynamic performance. For the unnormalized stack of pairwise errors, the optimum grows as $\beta\sqrt{N/2}$.

The threshold has a direct interpretation. The first follower can receive only the leader's packet. Every later follower needs at least two parents to attain the optimum; a single parent leaves an attack channel with unit incoming weight and a larger unavoidable physical error. One leader link plus two links for each remaining follower gives $1 + 2(N-2) = 2N-3$. The equal weights in (19) achieve this count without depending on aircraft-specific coefficients.

The required budget changes with the physical pairs used to measure formation error.

Corollary 1: If the physical graph is a tree directed away from the leader and its edges are available for communication, assigning weight one to each of these $N-1$ links attains the global lower bound $\Gamma_{\mathrm{ss}} = \beta/\sqrt{Q}$. This is the minimum feasible link budget.

Proof: An attack entering follower i propagates with unit weight to exactly its descendants. Thus Re_i is the indicator of the subtree rooted at i. Only the edge entering i crosses between this subtree and the remaining aircraft, so $\|HRe_i\|_2^2 = 1$. A possible first follower is adjacent to the leader and has $\rho_i = 1$, giving equality in (12). Leader reachability requires at least $N-1$ communication links. ■

E. Sparse Physical Graph Example

The general construction also applies when not every physical pair is present. Consider five aircraft with physical edges

$$\mathcal{E}_p = \{01, 02, 03, 04, 13, 14, 23, 24\},$$

where ji denotes $j \to i$. The graph has $Q = 8$ edges, and

$$K = \begin{bmatrix} 3 & 0 & -1 & -1 \\ 0 & 3 & -1 & -1 \\ -1 & -1 & 3 & 0 \\ -1 & -1 & 0 & 3 \end{bmatrix}. \tag{22}$$

Followers 1 and 2 can each be first in the common order and have the same $\rho_i = 15/7$. Choosing follower 1 gives

$$h = (1, 2/7, 3/7, 3/7)^T, \qquad \rho_1 = 15/7. \tag{23}$$

Lemma 1 therefore selects seven links with weights

$$\begin{aligned} w_{10} = 1, \quad & (w_{20}, w_{21}) = (5/7, 2/7), \\ & (w_{30}, w_{31}) = (4/7, 3/7), \\ & (w_{40}, w_{41}) = (4/7, 3/7). \end{aligned} \tag{24}$$

The residual terms in (15) are $75/49$, $48/49$, and $48/49$, all smaller than $\rho_1 = 105/49$. Thus this design attains the global steady-state lower bound

$$\Gamma_{\mathrm{ss}} = \beta\sqrt{15/56}. \tag{25}$$

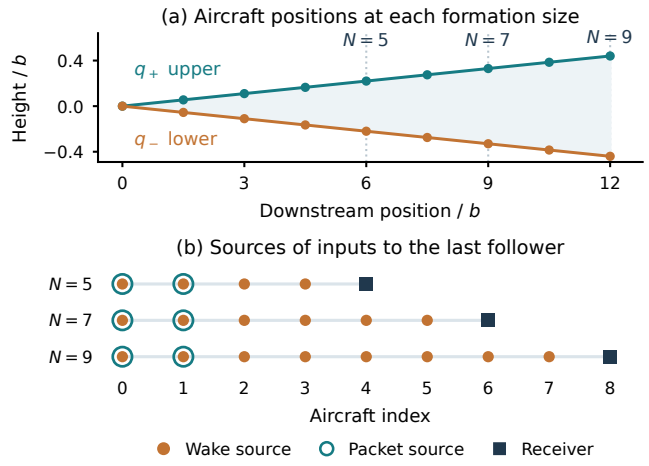


Fig. 3. Case-study geometries and inputs. (a) Dotted lines mark the last follower at each size. (b) Wake and communication inputs to the last follower with the links in equation (19).

TABLE I

LINEAR ATTACK GAINS WITH $2N-3$ LINKS. BOLD ENTRIES MARK THE LOWEST PEAK AMONG THE THREE LISTED DESIGNS FOR EACH SIZE AND GEOMETRY. SUPERSCRIPT U DENOTES A NUMERICAL UPPER BOUND. OTHER PEAKS ARE ATTAINED VALUES.

			Peak gain	
N	Design	Steady state	Upper q_+	Lower q_-
5	Equation (19)	9.1097	$\mathbf{9.1098}^{\mathrm{U}}$	13.9707
	Leader–predecessor	10.1849	10.1849	**11.1338**
	Two predecessors	9.5543	9.5543	13.4268
7	Equation (19)	7.4380	$\mathbf{7.4381}^{\mathrm{U}}$	17.7673
	Leader–predecessor	9.2859	9.2859	**9.5826**
	Two predecessors	7.8990	7.8990	15.3432
9	Equation (19)	6.4415	$\mathbf{6.4416}^{\mathrm{U}}$	23.4321
	Leader–predecessor	8.6053	8.6053	**8.6053**
	Two predecessors	6.8810	8.4183	17.8205

Seven links are also necessary. In every admissible order, the successive values of σ_i are $15/7, 7/3, 3, 3$. A six-link design has at least two followers with only one incoming link, including one after the first follower. Proposition 1 then gives $\Gamma_{\mathrm{ss}}^2 \geq (\beta^2/8)(7/3) > \beta^2(15/56)$. Thus this sparse graph has the same exact seven-link threshold as the five-aircraft all-pairs graph. The link $1 \to 2$ carries communication without a corresponding physical edge.

IV. Aircraft Case Studies

A. Aircraft Model and Geometries

We use a reduced cruise model of the NASA Lift+Cruise aircraft based on GUAM [18], with span-averaged wake loads based on [19]. Each aircraft has 18 states, including six controller states, cruises at 200 ft/s, and has wingspan $b = 14.478$ m.

We compare upper and lower echelons, denoted q_+ and q_-, for $N = 5, 7, 9$. They have the same downstream and lateral spacing and opposite height offsets (Fig. 3). For $i = 0, \ldots, N-1$, the position relative to the leader in downstream, lateral, and upward coordinates is

$$q_i^{\pm}/b = i(1.5,\ 1.029333,\ \pm 0.055). \tag{26}$$

In both geometries, every upstream aircraft contributes a wake with delay $\tau_{ji} = 0.35625(i-j)$ s, giving the physical graph in equation (18). We linearize about the operating points with these wakes and the same local controllers. All six operating points satisfy the clearance and aerodynamic limits of the model and have stable local dynamics.

At equilibrium, integral action balances constant wake loads. The weighted incoming packets determine the settled position offset, and velocity and course return to their nominal values. The same map H_0 holds for all six linearizations and preserves position-only offsets, as required by Assumption 1.

In feet, seconds, and radians, we use $S_a = \mathrm{diag}(0.1 I_6, 0.05\pi/180)$ and $W_F = \mathrm{diag}(I_3/0.095, I_3/2)$, giving $\beta \simeq 18.2194$. The reproducibility package [20] provides the aircraft and wake matrices, delays, and assembled closed-loop models.

B. Peak Attack Amplification

We compare three designs with $2N-3$ links each. Aircraft 1 receives only from the leader. Each later follower gives equal weight to two senders. Equation (19) uses the leader and aircraft 1. The alternatives use the leader and the follower's predecessor, or its two predecessors.

At each attack frequency, we maximize the gain over the link and the relative amplitudes and phases of the attack inputs. Table I compares the steady-state and peak gains. The Appendix gives the method for the full-frequency upper bounds.

With equation (19), both geometries attain the same steady-state bound. At q_+, the numerical upper bounds on peak gain lie within 0.0013% of this limit for all three sizes. At q_-, the gain peaks at a nonzero frequency (Fig. 4(a)). From five to nine aircraft, this peak rises from 13.97 to 23.43 as the steady-state gain falls from 9.11 to 6.44.

At q_-, the leader–predecessor design gives smaller attained peaks at all three sizes. Its five-aircraft upper bound is 11.14. Adjusting the weights on the seven links of equation (19) gives an upper bound of 10.87, more than 22% below the equal-weight peak of 13.97. The steady-state gain rises from 9.11 to 9.62.

The model gives equal upwash at the two heights. An upward displacement reduces it at q_+ and increases it at q_-. Swapping local wake derivatives in the five-aircraft models removes the observed peak at q_- and introduces one at q_+. Suppressing the delayed upstream wake terms also removes the peak at q_-. Both checks preserve the steady-state gain.

C. Communication Budget and Spacing

For five aircraft, we screen all forward link sets and locally optimize weights on selected candidates at budgets of four to ten links. Figure 5(a) compares the best designs found with the analytical budget bound. At q_+, seven links approach

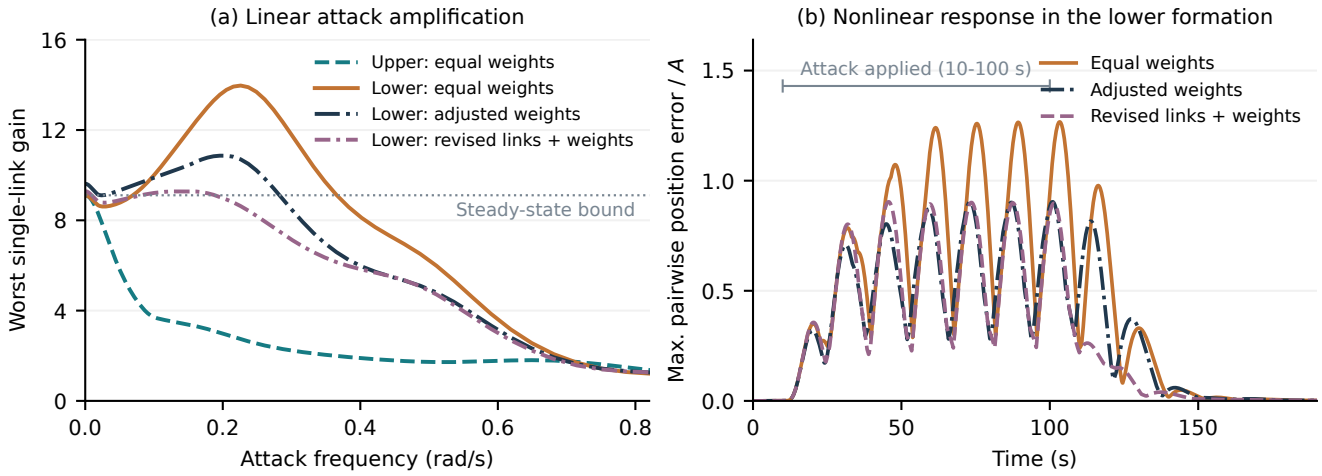


Fig. 4. Attack response for five aircraft. Equal weights uses equation (19); adjusted weights changes only those weights; revised links + weights also changes which seven links are used. (a) Both numerical designs reduce peak amplification in the lower formation, at the cost of a higher steady-state gain. (b) Under the same oscillating attack at q_-, both designs reduce the peak position error. The error is divided by the peak injected displacement A.

the steady-state limit. At q_-, numerical upper bounds are 9.30 with seven links and 9.111 with ten, compared with the limit 9.1097. The seven-link result is the revised design in Fig. 4. Additional links improve its dynamic gain although the steady-state optimum cannot decrease. Global dynamic optimality is not established.

We vary the lateral increment in (26) from $1.029333b$ to $1.35b$, holding the other increments, controllers, and weights fixed. All sampled formations pass the physical and local-stability checks. Figure 5(b) shows where the design ranking changes at q_-. The equal-weight peak falls from 13.97 to its steady-state value 9.11 by $1.04b$, whereas adjusted weights retain a steady-state gain of 9.62. The sampled peaks stay at their respective steady-state values for larger spacings. Thus weights adjusted for one geometry can lose their advantage at another.

D. Nonlinear Attack Responses

We corrupt vertical-position data on $0 \to 1$ from 10 to 100 s using a bias or an oscillation at 0.225 rad/s, near the linear peak at q_-. We first compare four seven-link designs at q_-, scaling each waveform to the same normalized energy $E_a = 20$. Figure 4(b) shows the oscillating attack on the equal-weight, adjusted-weight, and revised-link designs.

Adjusting weights reduces the maximum pairwise position error divided by the attack amplitude A from 1.267 to 0.904, a 29% reduction. The measured ratio $\|z\|_{L_2}/\|a\|_{L_2}$ over the simulation falls from 0.809 to 0.638, and to 0.519 for the revised-link design. The leader–predecessor design gives 0.541. These are responses to the common attack, rather than worst-case gains. For the energy-matched bias, adjusting weights increases this ratio from 0.544 to 0.566, illustrating the tradeoff with low-frequency performance.

All eight matched-energy simulations remain within the wake model's validity region. The reproducibility package also retains larger-amplitude tests, including cases that reach the wake-core clearance boundary.

V. Conclusion

We analyzed communication design against single-link attacks in aircraft formations with delayed wake coupling. We established a steady-state lower bound, constructed links and weights that can attain it, and determined minimum link requirements for selected physical graphs. Numerical and nonlinear studies showed how communication design can reduce attack amplification and how its benefits depend on formation geometry.

Future work will address coordinated multi-link attacks under a shared energy budget and seek analytical conditions connecting steady-state and dynamic optimality. Another direction is the joint design of formation geometry, communication links, and weights under range constraints and uncertainty in wake coupling and delays.

Appendix
Whole-Frequency Upper Bounds

After grounding the leader, collect equal wake delays and write

$$M(\omega) = \mathrm{i}\omega I - A - \sum_{\ell} e^{-\mathrm{i}\omega\tau_\ell} A_\ell, \qquad T_e(\mathrm{i}\omega) = CM(\omega)^{-1}B_e. \tag{27}$$

A diagonal state similarity transformation improves numerical scaling without changing the transfer. On an interval with midpoint ω_0 and half-width h, set $R_0 = M(\omega_0)^{-1}$ and $d_\ell = \min(2, \tau_\ell h)$. Define

$$\eta = h\|R_0\|_2 + \sum_{\ell} d_\ell \|R_0 A_\ell\|_F, \tag{28}$$

$$\kappa = h\|CR_0\|_2 + \sum_{\ell} d_\ell \|CR_0 A_\ell\|_F. \tag{29}$$

Here $\|\cdot\|_F$ denotes the Frobenius norm. The delayed-phase difference is bounded by d_ℓ. If $\eta < 1$, the resolvent identity and a Neumann-series bound give

$$\|T_e(\mathrm{i}\omega)\|_2 \le \|CR_0B_e\|_2 + \frac{\kappa\|R_0B_e\|_2}{1-\eta}. \tag{30}$$

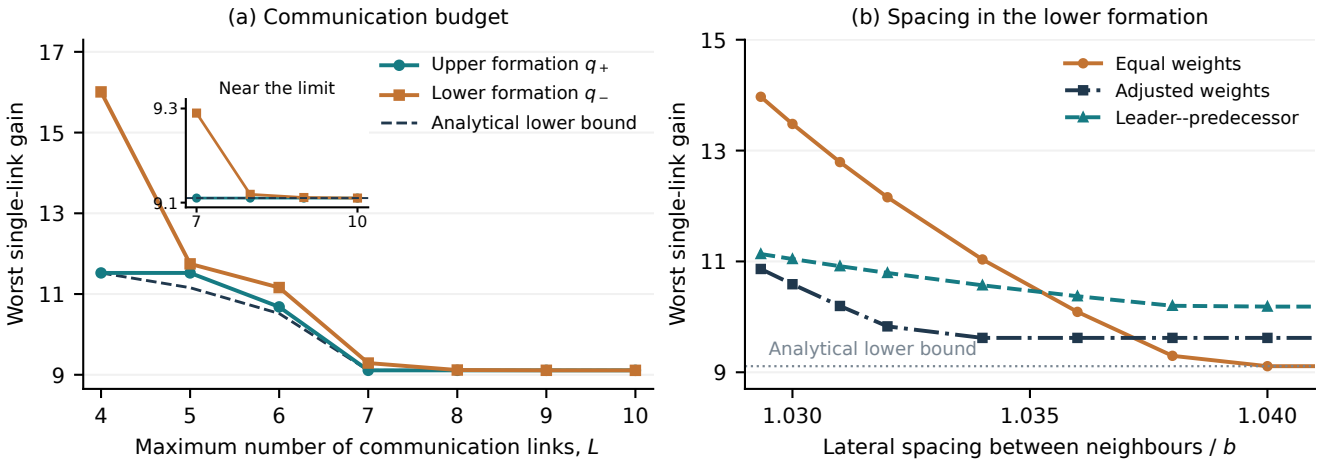


Fig. 5. Communication budget and lateral spacing in five-aircraft formations. Lower gains mean less attack amplification. (a) Each point shows the smallest gain found using at most L communication links. Allowing more links reduces the gains found, with a larger improvement for the lower formation q_-. The small plot provides an enlarged view of the differences near the analytical lower bound. (b) Three seven-link designs are compared in the lower formation q_-. Lateral spacing changes, while each design's links and weights remain fixed. Adjusted weights give the lowest gain at smaller spacings, whereas equal weights give the lowest gain at larger spacings.

Indeed, for $E = M(\omega) - M(\omega_0)$, $M(\omega)^{-1} - R_0 = -R_0 E(I + R_0 E)^{-1} R_0$, with $\|R_0 E\|_2 \le \eta$ and $\|C R_0 E\|_2 \le \kappa$. The same bound can be applied to a horizontal concatenation of link-input matrices, whose norm bounds that of every individual link.

For $a_0 = \|A\|_2 + \sum_\ell \|A_\ell\|_2$ and $\omega > a_0$, the tail satisfies

$$\|T_e(\mathrm{i}\omega)\|_2 \le \frac{\|C\|_2 \|B_e\|_2}{\omega - a_0}. \tag{31}$$

Intervals are subdivided until (30) meets the target; the stored cover and tail are checked for gaps. A sampled response above the target rejects that candidate bound. These inequalities are analytic; the reported upper values use padded floating-point evaluation rather than validated interval arithmetic. Stability is checked separately from the gain calculation.

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